NMOS Characterization

Analog Electronics Lab — Experiment 1

Field	Details
Student	Deva Prassad S
Roll Number	2025BEC0019
Programme	B.Tech Electronics and Communication Engineering
Institution	IIIT Kottayam
Tools	Cadence Virtuoso

1. Aim

To obtain the drain and transfer characteristics of an NMOS transistor and verify its DC operating point and region of operation, using Cadence Virtuoso.

2. Design Specifications

Parameter	Value
Device	nmos1v
Width (W)	120 nm
Length (L)	100 nm
Multiplier (m)	1
Supply voltage (VDD)	1.2 V
VGS sweep range	0 – 1.2 V
VDS sweep range	0 – 1.2 V
Technology	GPDK090

3. Circuit Description

The test schematic uses one nmos1v device with two independent DC sources: V0 at the gate sets VGS, and V1 at the drain sets VDS. Source and bulk are grounded, allowing VGS and VDS to be swept independently.

4. Simulation Procedure

1. Drew the schematic with independent VGS and VDS sources at the gate and drain.
2. Drain characteristics: swept V1 (VDS) from 0 to 1.2 V at fixed VGS and recorded ID.
3. Family of curves: used Parametric Analysis to sweep VGS (0–1.2 V, 5 steps), re-running the VDS sweep at each step.
4. Transfer characteristics: swept V0 (VGS) from 0 to 1.2 V at fixed VDS and recorded ID.

5. Results

5.1 Drain Characteristics (ID vs VDS)

ID rises with VDS and saturates once VDS exceeds VDSAT, consistent with the device entering saturation.

5.2 Family of ID–VDS Curves (VGS = 0, 0.3, 0.6, 0.9, 1.2 V)

Higher VGS gives proportionally higher saturation current, matching square-law MOSFET behavior.

5.3 Transfer Characteristics (ID vs VGS)

ID is negligible below threshold and rises with VGS above it, confirming correct turn-on behavior.

6. Observations

- ID follows the expected square-law dependence on VGS in saturation, and linear dependence on VDS in triode.
- The ID–VDS family of curves separates clearly by VGS, confirming gate control of channel current.

7. Conclusion

The drain and transfer characteristics of the NMOS device (GPDK090, W = 120 nm, L = 100 nm) were obtained using Cadence Virtuoso .